Chap 8 Worked probs

1) FLOPs for inf = FLOPs for forward pass

$= 2P = 810 \times 10^9$

$T_{math} = \frac{810 \times 10^9}{N \cdot C}$ where $C = 1.97 \times 10^{14}$ FLOPs/s

$T_{HBM} = \frac{2 \times 810 \times 10^9}{N \cdot W_{HBM}} \approx \frac{988}{N}$ ms

2) a) Model params: 8 Billion params in int8 = 8.0×10^9 bytes = 8GB

b) KV cache / token = 2LKH = 2(32)(8)(128) ≈ 65 kB / token

Total KV cache mem = BS (2LKH) ≈ 0.53 GB / sequence

With BS240 ⇒ total ≈ ~~129~~ 129 GB

c) peak working activations: [B,D] [B,F], [B,V]

A good approx. is to look at [B,V] ∵ V=128256,

∴ MEM logits = BV(2) = 2BV ≈ 61.6 MB

as an estimate for activations

d) Total Mem needed ≈ 8+129+0.1

≈ 137 GB

$N_{min} = \lceil \frac{137}{16} \rceil = 9$

∴ Smallest = 4×4

3) Model params ≈ 405 GB,

KV cache per token = 2LKH = 2(126)(8)(128) = 258 kB / tok

∴ KV cache for 128k seq = 33.82 GB

For Batch size of B, KV cache = (B·33.82) GB

∴ Total mem ≈ (405 + 33.82B) GB (activations are negligible)

Let B=1, then $N_{min} = \lceil \frac{438.82}{16} \rceil = 28$

∴ Minimum topology = 4×8

Speed considerations:

$T_{step} = T_{kv} + \max(T_{params}, T_{math})$

Let # TPU v5e $= n$

then $T_{params} = \dfrac{405 \times 10^9}{n(8.2 \times 10^{11})} = \dfrac{493.9}{n}$ ms

$T_{kv} = \dfrac{B(33.82 \times 10^9)}{n(8.2 \times 10^{11})} = \dfrac{41.25B}{n}$ ms

Each decode batch has roughly 2BP FLOPs

$T_{math} = \dfrac{2B(405 \times 10^9)}{n(1.97 \times 10^{14})} = \dfrac{4.11}{n} B$ ms

$T_{math} > T_{params}$ when $B \simeq 120$

$\therefore$ For $B < 120$, $T_{params} > T_{math}$, and $T_{step} = T_{kv} + T_{params} = \dfrac{41.25B + 493.9}{n}$ ms

$\dfrac{493.9 + 41.25B}{n} \le 15$ ms

For a given n, we find the largest possible B to fit the inequality

n	B_{mem}	$B_{latency}$	B_{best}	Step time (ms)
32	3	0	—	—
64	18	11	11	14.81
128	48	34	34	14.81
256	109	81	81	14.98

Working, $n = 32$, $B_{mem} \le \dfrac{\text{total HBM} - \text{model weights}}{\text{KV cache per seq}} \Rightarrow B_{mem} = \left\lfloor \dfrac{64(16) - 405}{33.82} \right\rfloor = 18$

- Calculate throughput $= \dfrac{B}{T_{step}}$

n	Throughput (tokens/s)
64	~743
128	~2295
256	~5407

$\Rightarrow$ 16x16 has highest throughput //

• ICI comms:

$$P_{max} \simeq \frac{F}{8B}$$ for ordinary decode tensor parallelism.

~~for~~ Llama 405B, $F = 53248$ and at $B = 81$,

$$P_{max} = \frac{53248}{8(81)} \simeq 82 \Rightarrow 256 \gg 82 \Rightarrow$$ comms bound.

However, testing ~~the~~ $n = 128$, $B = 34$

$$P_{max} = \frac{53248}{8(34)} \simeq 196 > 128, \Rightarrow$$ a simpler and safer config

Theoretical min step time, at $n = 256$ and $B = 1$

$$T_{params} \simeq \frac{493.9}{256} \simeq 1.93\,ms$$

$$T_{kv} = \frac{41.25}{256} \simeq 0.161\,ms$$

$$T_{math} = \frac{4.11}{256} \simeq 0.016\,ms$$

$$\therefore T_{min\,step} = 1.93 + 0.161 \simeq 2.09\,ms/token$$